Supplement A

Smooth Registry Realizations of Lie Groups

A Rigorous Extension of the Structural Registry Framework

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This supplement preserves ontological hygiene: established differential-geometric facts are stated as theorems or corollaries; stronger claims about finite polygonal realization, Boy-surface encoding, or universal Hopf generation are explicitly retained as conjectural research directions.

Contents

Abstract

This supplement extends the Structural Registry framework from discrete polygonal and polyhedral registries to smooth manifolds supporting Lie group structures. The objective is not to replace classical Lie theory, but to show that the registry formalism naturally accommodates smooth group actions, infinitesimal generators, maximal tori, Weyl symmetries, and root decompositions. The principal result establishes that every finite-dimensional Lie group admits a faithful smooth registry realization: group elements act as registry transport morphisms, and the Lie algebra is recovered from infinitesimal transport fields. Stronger claims regarding universal finite polygonal generation, Boy-surface realization of brackets, or Hopf-fibration generation of all compact Lie groups remain explicitly conjectural.

Keywords

Structural registry; Lie group; Lie algebra; smooth registry; registry transport; left translation; infinitesimal generators; maximal torus; Weyl group; root decomposition; Hopf fibration; SU(2).

1 Motivation

The foundational Structural Registry manuscript develops a framework in which nested polygonal and polyhedral incidence structures generate scale-invariant ratio registries. In that setting, incidence precedes measurement; measurement precedes ratio; and ratio precedes harmonic interpretation.

The present supplement asks whether the same registry language extends from discrete incidence complexes to smooth geometric structures. Lie groups provide a natural test case because they already unify geometry, transport, symmetry, and infinitesimal algebra. The guiding question becomes:

> Can a Lie group be represented as a smooth registry whose transport morphisms recover both the group law and the associated Lie algebra?

The answer, with a properly defined smooth registry, is yes. The proof uses standard Lie theory: every finite-dimensional Lie group acts faithfully on itself by left translation, and its Lie algebra is recovered as the space of left-invariant vector fields under commutator.

2 Notation

Symbol	Meaning
G	finite-dimensional Lie group
e	identity element of G
$\mathfrak{g}$	Lie algebra of G, identified with T_eG
L_g	left translation map $h \mapsto gh$
$\mathrm{Diff}(G)$	diffeomorphism group of the smooth manifold G
$\widetilde{X}$	left-invariant vector field generated by $X \in T_eG$
$[X, Y]$	Lie bracket in $\mathfrak{g}$
$T \subseteq G$	maximal torus of a compact connected Lie group
$W = N_G(T)/T$	Weyl group
Φ	root system
$\mathcal{R}_G$	smooth registry associated with G
$\mathcal{T}$	family of registry transport morphisms

3 Smooth Registry Objects

The discrete registry in the foundational manuscript may be written as an incidence complex

$$\mathcal{R} = (V, E, F, C, \iota), \tag{1}$$

where V denotes vertices, E edges, F faces, C higher-dimensional cells, and ι the incidence relation. For Lie groups, the discrete registry must be extended to a smooth setting.

Definition 3.1 (Smooth registry)**.** Let G be a smooth manifold. A smooth registry modeled on G is a tuple

$$\mathcal{R}_G = (O, G, \mathcal{F}, \mathcal{M}, \mathcal{I}, \mathcal{T}), \tag{2}$$

where:

(i) O is a distinguished registry origin, identified with a chosen base point; for Lie groups this will be the identity e;

(ii) G is the smooth registry manifold;

(iii) $\mathcal{F}$ is a chosen layer structure, such as a foliation, fibration, torus decomposition, cell decomposition, or nested atlas;

(iv) $\mathcal{M}$ is a family of geometric or differential measurements associated with G and $\mathcal{F}$;

(v) $\mathcal{I}$ is the corresponding family of invariant ratios or normalized relational quantities;

(vi) $\mathcal{T}$ is a family of smooth registry transport morphisms.

Remark 3.2. This definition does not require a Lie group to arise from finite polygonal or polyhedral data. The smooth registry generalizes the incidence-first philosophy without claiming that all smooth structures reduce to finite combinatorics.

4 Registry Transport

Definition 4.1 (Registry transport)**.** Let $\mathcal{R}_G$ be a smooth registry. A registry transport is a smooth map

$$T : \mathcal{R}_G \longrightarrow \mathcal{R}_G \tag{3}$$

that preserves the registry structure under consideration. In the Lie group case, the relevant preserved structure is smooth transport of the registry manifold together with the distinguished identity-based infinitesimal structure.

Definition 4.2 (Faithful transport realization)**.** A Lie group G admits a faithful smooth registry realization if there exists an injective group homomorphism

$$\Theta : G \hookrightarrow \mathrm{Diff}(G) \tag{4}$$

whose image consists of registry transport morphisms.

5 Faithful Smooth Registry Realization

Theorem 5.1 (Faithful Smooth Registry Realization of Lie Groups)**.** *Let G be a finite-dimensional Lie group. Then G admits a faithful smooth registry realization in which group elements act as registry transport morphisms and the Lie algebra $\mathfrak{g}$ is recovered from infinitesimal transport fields.*

Proof. Take the registry manifold to be the smooth manifold underlying G, and choose the distinguished origin to be the identity element $e \in G$. For each $g \in G$, define

$$T_g = L_g, \tag{5}$$

where left translation is

$$L_g(h) = gh. \tag{6}$$

For $g, h \in G$,

$$L_g \circ L_h = L_{gh}. \tag{7}$$

Therefore the assignment

$$\Theta : G \longrightarrow \mathrm{Diff}(G), \qquad \Theta(g) = L_g, \tag{8}$$

forms a group homomorphism.

This homomorphism is injective. If $\Theta(g) = \Theta(e)$, then $L_g = L_e$. Evaluating both maps at e gives

$$g = L_g(e) = L_e(e) = e. \tag{9}$$

Hence Θ is faithful.

The tangent space at the identity,

$$T_eG, \tag{10}$$

is the Lie algebra $\mathfrak{g}$. Each $X \in T_eG$ extends uniquely to a left-invariant vector field $\widetilde{X}$ on G by

$$\widetilde{X}(g) = d(L_g)_e X. \tag{11}$$

The commutator of two left-invariant vector fields is again left-invariant. The Lie bracket on $\mathfrak{g}$ is recovered by evaluating the commutator at the identity:

$$[X, Y]_{\mathfrak{g}} = [\widetilde{X}, \widetilde{Y}](e). \tag{12}$$

Thus the global group action is represented by registry transports, while the Lie algebra is recovered from infinitesimal registry transports. This proves the claim. $\square$

Corollary 5.2 (Lie bracket as infinitesimal registry commutator)**.** *In the smooth registry realization of a finite-dimensional Lie group, the Lie bracket is the commutator of infinitesimal registry transport fields.*

Proof. By the preceding theorem, each $X \in \mathfrak{g}$ determines a left-invariant infinitesimal transport field $\widetilde{X}$. The ordinary vector-field commutator $[\widetilde{X}, \widetilde{Y}]$ remains left-invariant and determines the bracket $[X, Y]$ at the identity. Hence the Lie bracket is recovered as the commutator of infinitesimal registry transports. □

6 Compact Lie Groups and Nested Torus Registries

For compact connected Lie groups, the smooth registry admits a more explicitly nested description through maximal tori and root decompositions.

Theorem 6.1 (Compact Lie Group Nested Registry)**.** *Let G be a compact connected Lie group. Then any maximal torus $T \subseteq G$ determines a nested angular registry inside the smooth registry realization of G. The Weyl group acts as a registry symmetry, and the root decomposition of $\mathfrak{g}_{\mathbb{C}}$ decomposes infinitesimal transport directions relative to this torus.*

Proof. Let $T \subseteq G$ be a maximal torus. Since T is compact, connected, and abelian, it is isomorphic to a finite product of circles:

$$T \cong (S^1)^r, \tag{13}$$

where r is the rank of G. Thus T supplies a nested angular registry consisting of commuting circle directions.

The normalizer $N_G(T)$ acts on T by conjugation. The quotient

$$W = N_G(T)/T \tag{14}$$

defines the Weyl group, which acts as a finite symmetry group of the torus registry.

For the complexified Lie algebra, the standard root decomposition is

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha. \tag{15}$$

Here $\mathfrak{t}_{\mathbb{C}}$ is the complexified Lie algebra of the maximal torus, Φ is the root system, and each $\mathfrak{g}_\alpha$ is a root space. This decomposes infinitesimal transport directions according to their response to torus action. Hence the compact connected Lie group carries a nested torus-root registry refining the smooth registry realization. □

Remark 6.2. This theorem does not claim that every compact Lie group is generated by finite polygons. It says that the standard maximal-torus and root decomposition of compact Lie theory can be interpreted as a nested smooth registry refinement.

7 Example: SU(2) and the Hopf Registry

The group $SU(2)$ provides the canonical example.

Proposition 7.1 (SU(2) as a smooth registry)**.** *The compact Lie group $SU(2)$ is diffeomorphic to S^3 and admits a smooth registry compatible with the Hopf fibration*

$$S^1 \hookrightarrow S^3 \longrightarrow S^2. \tag{16}$$

Proof. Identify $SU(2)$ with the unit quaternions, equivalently with the unit sphere in $\mathbb{C}^2$:

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}. \tag{17}$$

The Hopf map may be written as

$$\pi : S^3 \to S^2, \qquad (z_1, z_2) \mapsto \left(2z_1\overline{z}_2, |z_1|^2 - |z_2|^2\right), \tag{18}$$

where the first coordinate is interpreted as a complex coordinate on the equatorial plane of S^2 and the second as the vertical coordinate. Each fiber is a circle S^1. Thus $SU(2) \cong S^3$ carries a natural nested registry with circular fibers over a spherical base. □

Corollary 7.2 (Registry realization of $\mathfrak{su}(2)$)**.** *The Lie algebra $\mathfrak{su}(2)$ is recovered from infinitesimal registry transports on S^3 by left-invariant vector fields.*

Proof. This is the specialization of Theorem 5.1 to $G = SU(2)$. Since $SU(2)$ acts on itself by left multiplication, and $SU(2) \cong S^3$, the infinitesimal transports of the smooth registry recover $\mathfrak{su}(2)$ through the commutator of left-invariant vector fields. □

8 Relation to the Discrete Structural Registry

The foundational Structural Registry paper establishes a discrete pathway:

$$\text{incidence} \longrightarrow \text{measurement} \longrightarrow \text{ratio} \longrightarrow \text{interpretation}. \tag{19}$$

The present supplement establishes the smooth analogue:

$$\text{smooth manifold} \longrightarrow \text{transport} \longrightarrow \text{infinitesimal field} \longrightarrow \text{Lie algebra}. \tag{20}$$

These two pathways are compatible rather than identical. Finite polygonal and polyhedral registries provide discrete constructive submodels, approximants, or symmetry skeletons. Smooth Lie registries provide continuous transport structures. The claim established here is not that one reduces without remainder to the other; the established claim is that the registry formalism can faithfully represent both.

9 Ontological Hygiene

9.1 Established within this supplement

The following claims are established by the definitions and theorems above:

1. Every finite-dimensional Lie group admits a faithful smooth registry realization by left translation.
2. The associated Lie algebra is recovered from infinitesimal registry transports via left-invariant vector fields.
3. Compact connected Lie groups admit nested torus-root registry refinements through maximal tori, Weyl groups, and root decompositions.
4. $SU(2) \cong S^3$ provides a canonical smooth registry compatible with the Hopf fibration.

9.2 Conjectural extensions

The following stronger statements are not proved here and should remain conjectural unless separately established:

1. Every compact Lie group arises from finite polygonal or polyhedral incidence alone.
2. Every root system admits a purely polygonal constructive realization.
3. Boy-surface singularities directly measure Lie brackets.
4. Hopf fibrations alone generate all compact Lie groups.
5. The category of compact Lie groups is equivalent to a category of finite nested geometry registries.

Conjecture 9.1 (Finite registry approximation)**.** Compact Lie group registries may admit increasingly faithful finite polygonal, polyhedral, or incidence-complex approximations preserving selected transport and symmetry data.

Conjecture 9.2 (Higher registry transport)**.** Fibrations such as the Hopf fibration may represent higher-order registry transports between smooth incidence layers and discrete registry approximants.

10 Conclusion

This supplement establishes a rigorous bridge from Structural Registry theory to standard Lie theory. Every finite-dimensional Lie group admits a faithful smooth registry realization by left translation. Its Lie algebra is recovered from infinitesimal registry transports through left-invariant vector fields and their commutators. Compact connected Lie groups further admit nested torus-root registry refinements through the standard machinery of maximal tori, Weyl groups, and root decompositions.

The result is neither a replacement for Lie theory nor an unsupported claim that all Lie groups arise from finite polygons. It is a precise compatibility theorem: the registry formalism can represent the global and infinitesimal structure of Lie groups while preserving its own constructive emphasis on origin, layering, transport, and invariant relation.

Consequently, smooth registry theory provides a disciplined extension of the discrete nested-geometry framework and supplies a mathematically sound foundation for later work on resonance registries, higher transport structures, and possible finite approximations of continuous symmetry.

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